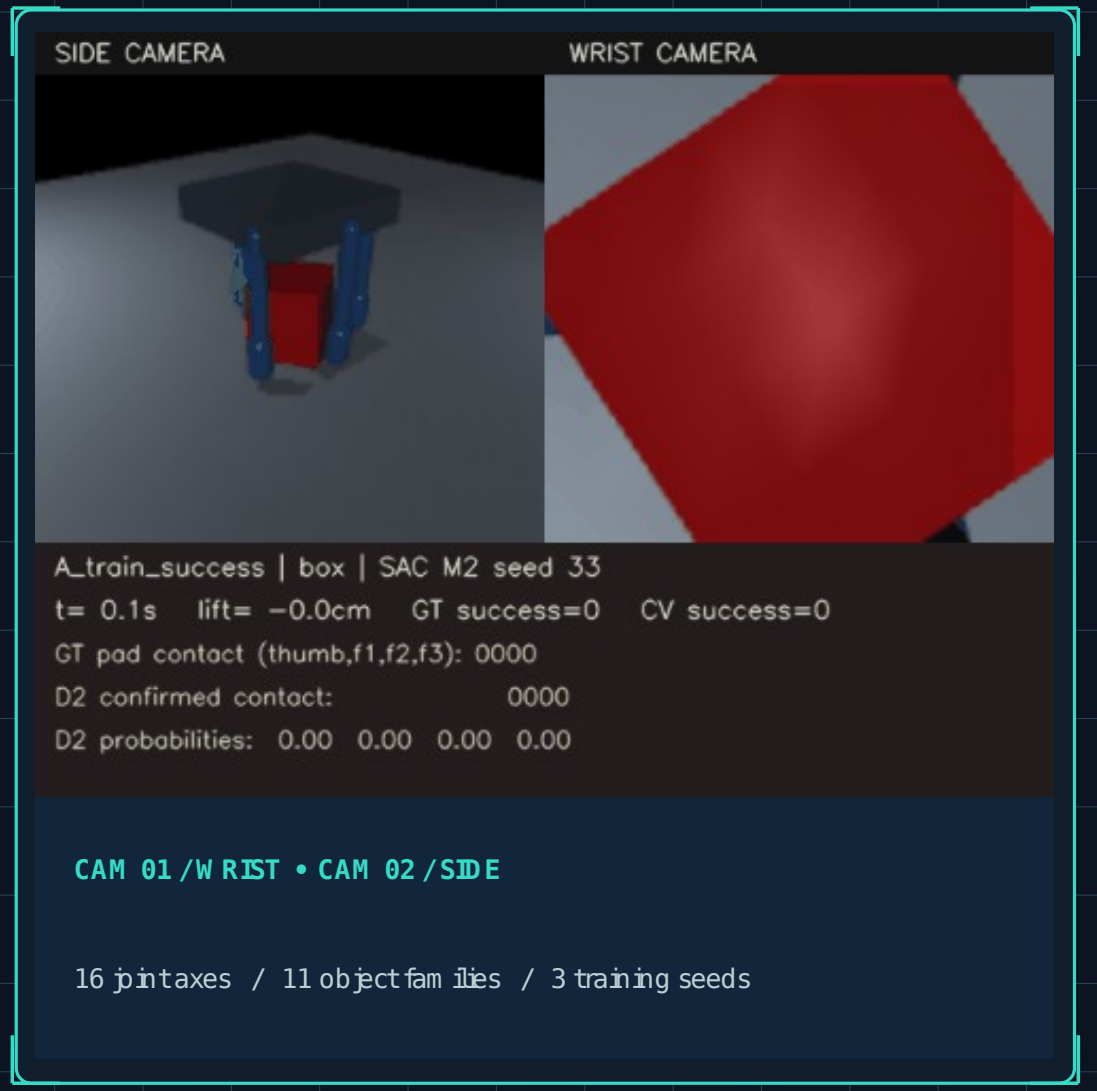
TOUCH IS A VISUAL PROBLEM.

Multi-finger grasping with deep reinforcement learning and vision-confirmed contact.

MuJoCo + D2CNN-GRU + SAC



THE ROBOT NEVER GETS A TOUCH BUTTON

In M2, contact enters reward and observation through images, not privileged simulator flags.

01 **SIMULATOR**
hand + object

›

02 **RGB-D**
wrist / side

›

03 **D2 DETECTOR**
temporal CNN–GRU

›

04 **CONFIRM**
K-frame filtering

›

05 **SAC ACTOR**
16 or 19 controls

M0 / ORACLE Exact simulator contact: comparison only.

M2 / VISION D2 contact flags + confidence: deployed signal.

GT contact remains reserved for labels and independent evaluation.

THE HAND. THE OBJECTS. THE RULE.

LEVEL A

16 finger actions

No scripted closing.
Scripted lifting after grasp.

LEVEL B

19 finger + wrist actions

Scripted descent / closure / lift.
RL learns residual corrections.

SUCCESS IS NOT JUST LIFT

10 cm	height above rest
3 s	sustained hold
2 pads	including thumb

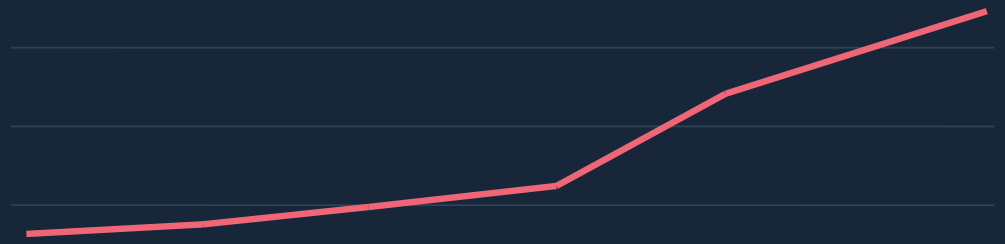
WHY ONE-STEP SAC FAILED

Early policies learned a grasp — and then forgot it as critic estimates inflated.

V3.1 / ONE-STEP

~34,000

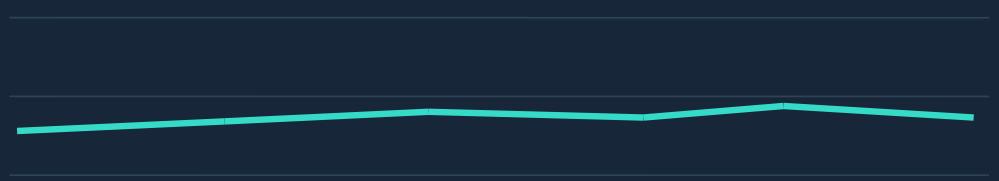
estimated Q by 195k steps



V3.3 / 10-STEP REPLAY

~25–40

logged Q values across final runs

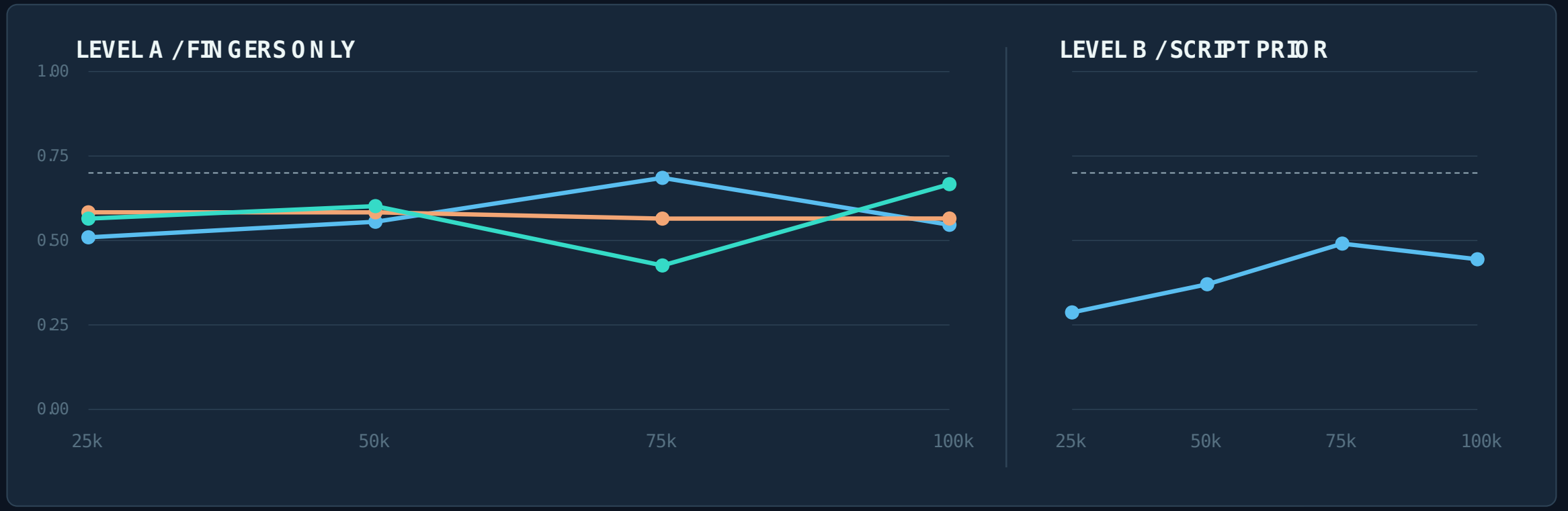


LayerNorm + n-step = 10

Observed association, not an isolated causal ablation.

A LEARNED GRASP IS NOT A MONOTONIC CURVE

Selection success / 36 fixed episodes at each checkpoint. Three independent seeds.



100k steps / seed | 3 seeds | 12 completed runs

THE INDEPENDENT TEST RESETS THE STORY

Nominal / ground-truth success. The scripted controller still wins.



+18.7 pp

M2 over random
Level A / nominal

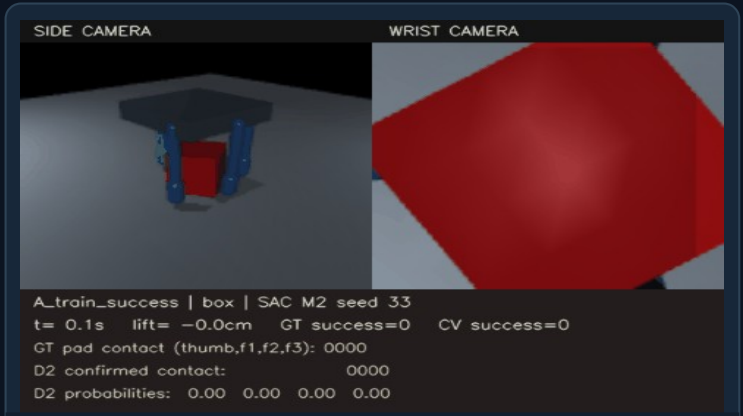
−20.3 pp

CV vs GT gap
M2 on unseen shapes

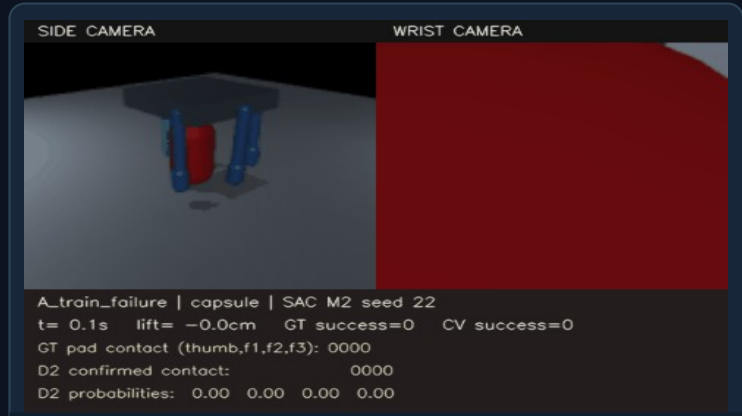
LEVEL B / M2 = 50.9% · OPEN-LOOP = 52.8% · STEERING SERVO = 87.2%

WHAT THE ROBOT LEARNED. WHAT IT DID NOT.

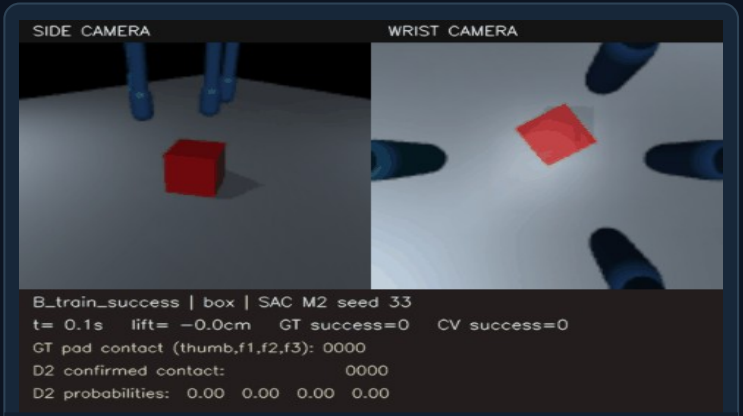
Replay video snippets for the committee; scene stills below are from the first captured frame.



A / SUCCESS ▶ MP4



A / FAILURE ▶ MP4



B / SUCCESS ▶ MP4

WORKED

Independent Level-A M2: 62.6% GT; 70.3% on unseen procedural families.

LIMITED

M2 did not outperform Blind or the strong scripted controllers.

NEXT

Improve detector robustness and adaptive wrist control; add real-world YCB evaluation.

VIDEOS ARE EXTERNAL MP4 FILES: attach them to the final submission package.